

Provider Costing

FarmVaidya Voice Agent - STT / LLM / TTS / Infrastructure / Concurrency

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USD to INR rate used:95.28 (project's own USD_TO_INR_RATE, .env)

- Green rows = currently active provider (.env)
- Orange rows = configured/available but no published price or limit

Rates below are the provider's own published list price, pulled from this project's own .env comments where already verified (dated), or looked up fresh where not. Several providers (Svara/Kenpath, Gnani.ai) do not publish list pricing at all - they quote custom/enterprise rates on request - those are flagged rather than guessed. There is no separate "Azure DB" in this project: Postgres runs locally on the dev Windows PC. The only real Azure resource is the VM (sethu-admin1 / agmarknet-scraper-vm) that runs the market-price scraper - its cost is in the Infrastructure section. Section 5 covers concurrent-call / concurrency limits, which is NOT documented anywhere in .env or the codebase - it was pulled live from each provider's account where an API for it exists, and flagged as unknown where it doesn't.

1. Speech-to-Text (STT)

Provider	Model	Native rate	INR / min	Status
Soniox	stt-rt-v5	\$0.12/hr = \$0.002/min	Rs 0.1906	ACTIVE
Sarvam AI	saaras:v3	Rs 30/hour	Rs 0.50	configured
Gnani.ai	Prisma v2.5	not published	custom quote	configured

STT_PROVIDER=soniox is currently active (Bot.py startup log: "STT: SONIOX model=stt-rt-v5"). Soniox is also by far the cheapest of the three published rates.

Source: Soniox rate is this project's own .env comment, verified against soniox.com/pricing on 2026-08-07. Sarvam rate from sarvam.ai/api-pricing. Gnani.ai has no public pricing page - enterprise/custom quote only, contact required.

2. LLM

Provider	Model	In \$/1M tok	Out \$/1M tok	Status
Google Vertex AI	gemini-2.5-flash	\$0.30	\$2.50	ACTIVE
OpenAI	gpt-4o	\$2.50	\$10.00	configured
Groq	llama-3.3-70b-versatile	\$0.59	\$0.79	configured

Provider	Model	In Rs/1M tok	Out Rs/1M tok	
Google Vertex AI	gemini-2.5-flash	Rs 28.58	Rs 238.20	
OpenAI	gpt-4o	Rs 238.20	Rs 952.80	
Groq	llama-3.3-70b-versatile	Rs 56.22	Rs 75.27	

LLM_PROVIDER=vertex is currently active. Vertex/Gemini is billed through Google Vertex AI, not the Gemini API directly - this project's .env explicitly notes that parity with Gemini API pricing "hasn't been separately verified against the Vertex AI pricing page," so treat this as the best public estimate, not a confirmed Vertex-specific rate. Groq is far cheaper per token but isn't the active choice (Vertex was picked for quality/latency, not cost).

3. Text-to-Speech (TTS)

Provider	Model	Native rate	INR / min	Status
Svara (Kenpath)	svara-1	not published	custom quote	ACTIVE
Soniox	tts-rt-v2	\$0.70/hr = \$0.0117/min	Rs 1.11	configured
Murf AI	Falcon-2	\$0.01 / 1,000 chars	~Rs 0.95 (approx)*	configured
Cartesia	sonic-3.5	\$0.065 / 1,000 chars	~Rs 5.57 (approx)*	configured
Bakbak (Raya)	standard	Rs 0.39/min (billed)	Rs 0.39	configured
Sarvam AI	bulbul:v3	Rs 30 / 10,000 chars	~Rs 2.70 (approx)*	configured
xAI / Grok	grok-tts	\$15 / 1,000,000 chars	~Rs 1.29 (approx)*	configured

* Char-billed providers (Murf, Cartesia, Sarvam, xAI) don't have a fixed Rs/min - the INR/min column is a rough approximation assuming ~150 spoken WORDS per minute (roughly 900 characters at ~6 chars/word incl. spaces; varies a lot by language and speaking speed), converted at Rs 95.28/\$. Murf's own figure is quoted directly from their stated "1 cent per minute" equivalence rather than this project's word-rate assumption. Soniox TTS and Bakbak are billed per actual audio-minute instead, which is what this project's own cost tracker (bot_processors/voice/latency.py) is built around.

TTS_PROVIDER=svara is currently active, but Svara/Kenpath Labs does not publish list pricing anywhere (docs.kenpathlabs.com/rate-limits: "email hello@kenpathlabs.com with your expected concurrency and monthly character volume"). Live response headers from this account (x-ratelimit-remaining-*) confirm real quota, not a free-tier account, but no INR rate can be quoted without that custom quote.

Known issue: .env's TTS_PRICE_PER_MIN is still set to Rs 1.11 (Soniox TTS's rate) from before the switch to Svara - the live call-end cost summary log line will show the wrong Rs/min against real Svara usage until this is corrected with Svara's actual quoted rate.

4. Infrastructure

Resource	Spec	USD	INR
Azure VM (compute)	Standard_B2als_v2, Central India	\$0.0246/hr	Rs 2.34/hr
	2 vCPU / 4 GiB, Ubuntu 24.04	\$17.96/mo	Rs 1,711/mo
Postgres ("Azure DB")	Runs locally, NOT on Azure	\$0	Rs 0

sethu-admin1 / agmarknet-scraper-vm runs the market-price scraper daemon (bot_processors/pricing/scraper_daemon.py) as an always-on systemd service. This is the ONLY Azure resource in the project - there is no Azure-hosted database. Postgres (the farmvaidya DB the voice bot and scraper both write to) runs locally on the dev Windows PC and is reached from the VM over a Tailscale tunnel, at zero additional Azure cost.

Azure VM rate: pay-as-you-go, Linux, no reserved-instance discount applied. Excludes OS disk/storage and bandwidth, which are small (typically a few dollars/month for a 30GB standard disk) but not itemized here. Source: azurespeed.com/AzureVmPricing/Standard_B2als_v2, checked 2026-08-17.

5. Concurrent Call Limits

Not documented anywhere in .env or the codebase. Pulled live from each provider's own account/API where possible; flagged "unknown" where no such API exists.

Layer	Concurrency limit	Verified how	Cost if exceeded	Status
Soniox STT	10 concurrent (org)	live API call	429 rejected	VERIFIED
Svara TTS	unknown total; 7 streams remaining	live headers only	429 rejected	PARTIAL
Vertex AI LLM	unknown (quota API disabled)	attempted, blocked	429/latency spike	UNKNOWN
Postgres pool	10 connections (db_pool.py)	db_pool.py	queued, not dropped	VERIFIED
Tata Smartflo	unknown - account/channel-specific	not published	call fails to connect	UNKNOWN

Soniox: GET <https://api.soniox.com/v1/concurrency-limits>, live-queried against this project's real SONIOX_API_KEY on 2026-08-17. Response: project limits are null (unlimited, bounded by org), organization limits = 10 concurrent transcribe sessions, 3 concurrent TTS streams, 10 "voice agent" sessions. Since every live call holds one dedicated STT WebSocket, 10 is a hard ceiling on simultaneous calls system-wide.

Svara: the hosted API only exposes remaining budget in response headers (x-ratelimit-remaining-requests/streams/characters), never the total limit. Live-queried just now: 99 requests/min remaining, 7 streams remaining, ~500,000 characters/month remaining - clearly a paid tier (free tier caps at 10 req/min / 2 streams / 10K chars-month), but the exact ceiling isn't obtainable via the API. Check [platform.kenpathlabs.com's billing/plan page](https://platform.kenpathlabs.com/billing/plan) for the real number.

Vertex AI: attempted a live query via Google's Cloud Quotas API (cloudquotas.googleapis.com) using this project's own service-account.json credentials - blocked with a 403 SERVICE_DISABLED error, because that API isn't enabled on vertex-ai-project-494805. Publicly, Gemini 2.5 Flash's commonly-cited default is 10 RPM / 250K TPM, but that figure is the AI-Studio/free-tier number, not necessarily what this billed Vertex project gets (Vertex uses Dynamic Shared Quota, usually higher for paying projects). Enable Cloud Quotas API on the project to get a real number.

Tata Smartflo: not found in the codebase, not in Tata's public API docs, not in their Voice Streaming SOP page. Smartflo concurrency is licensed per-channel on the specific account (however many channels were purchased), not a fixed platform-wide number Tata publishes anywhere. No API-key-based way to query this remotely was found - it requires checking the Smartflo dashboard directly or contacting Tata support. This also determines whether inbound and outbound calls share one channel pool or have separate limits - undocumented either way.

Bottom line: the real ceiling on concurrent calls today is the MINIMUM across this chain. Soniox's 10-concurrent-session cap is the only fully confirmed hard number in the AI stack. Tata Smartflo's channel count is the biggest unknown and the most likely actual bottleneck for real call volume - check that first.